EXPLORATORY DATA ANALYSIS (EDA) PROJECT

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INTRODUCTION

The Ecommerce Data Set for Exploratory Data Analysis (EDA) project aims to unravel the ever-dynamic field of internet product sales using the Ecommerce dataset. The dataset includes a rich array of numerical and categorical variables, such as product pricing, customer ratings, publication dates, and various other attributes. This comprehensive dataset serves as a foundation for a deep dive into the intricacies of online commerce, offering insights into factors influencing customer behavior, product popularity, and overall market dynamics.

The motivation behind this project lies in the increasingly vast and dynamic nature of the internet marketplace. Companies often struggle to identify which factors have the greatest impact on sales and to determine how marketing and pricing strategies influence customer engagement and purchase decisions. This EDA project aims to address these challenges by employing statistical analysis, visualization techniques, and data summarization methods to extract meaningful insights from the dataset. The ultimate goal is to provide valuable recommendations for businesses and stakeholders to optimize their marketing efforts and improve their overall performance in the online marketplace.

The analysis follows a systematic step-by-step approach. It starts with an initial examination of the dataset using descriptive statistics and visual summaries, followed by data cleaning, treatment of missing values, and investigation of outliers. The subsequent phases delve into univariate, bivariate, and multivariate analysis, exploring the relationships between different variables and their impact on sales. Hypothesis testing will be conducted to validate the observed patterns and trends, ensuring the robustness of the insights. Finally, the project culminates in a comprehensive report that synthesizes the findings and provides actionable recommendations for businesses to enhance their online sales strategies.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
from sklearn.model_selection import train_test_split
```

```
df = pd.read_csv('/kaggle/input/ecommerce-data/data.csv')
df.head(10)
```

	Order	CompanyId	Gender	Monthly Income	Population	Price	Rate	Discount	Profit	Quantity
0	1145	10121	Male	25	2890	25	Good	10	15	Two
1	1146	10122	Male	50	2890	25	Good	10	15	Two
2	1147	10123	Male	75	2890	25	Good	10	15	Two
3	1148	10124	Male	100	2890	25	Good	10	15	Two
4	1149	10125	Male	0	2890	25	Good	10	15	Two
5	1150	10126	Male	25	2890	25	Good	10	15	Two
6	1151	10127	Male	50	2890	25	Good	10	15	Two
7	1152	10128	Male	75	2890	25	Good	10	15	Two
8	1153	10129	Male	100	2890	25	Good	10	15	Two
9	1154	10130	Male	0	2890	25	Good	10	15	Two

We have searched the world's literature on creative industry related keywords.

1. Descriptive statistics.

```
df.describe()
```

	Order	CompanyId	Monthly Income	Population	Price	Rate	Discount
count	99.000000	99.000000	99.000000	99.000000	99.000000	99.000000	99.000000
mean	1195.000	10.242424	50.000000	3.030303	28.333333	10.750000	15.333333
std	29.010342	10.242424	29.010342	3.030303	17.320508	2.217356	5.291502
min	1145.000	10.000000	0.000000	2.000000	10.000000	10.000000	10.000000
25%	1170.000	10.000000	25.000000	2.000000	25.000000	10.000000	10.000000
50%	1195.000	10.000000	50.000000	2.000000	25.000000	10.000000	15.000000
75%	1220.000	10.000000	75.000000	4.000000	25.000000	10.000000	20.000000
max	1245.000	10.000000	100.000000	5.000000	50.000000	12.000000	20.000000

```
df.info()
```

```
<class RangeIndex>
RangeIndex: 99 entries, 0 to 98
Data columns (total 10 columns):
 Order          int64
 CompanyId      int64
 Gender         object
 Monthly Income int64
 Population     int64
 Price          int64
 Rate           object
 Discount       int64
 Profit         int64
 Quantity       object
dtype: int64, object: 2
memory usage: 7.8+ KB
```

2. Dealing with Null/Missing Values.

```
df.isnull().sum()
```

```
Order          0
CompanyId      0
Gender         0
Monthly Income 0
Population     0
Price          0
Rate           0
Discount       0
Profit         0
Quantity       0
dtype: int64
```

```
df.isna().sum()
```

```
Order          0
CompanyId      0
Gender         0
Monthly Income 0
Population     0
Price          0
Rate           0
Discount       0
Profit         0
Quantity       0
dtype: int64
```

```
df.duplicated().sum
```

```
0
```

```
df.nunique()
```

```
Order          99
CompanyId       2
Gender          2
Monthly Income  2
Population      4
Price           2
Rate            2
Discount        2
Profit          2
Quantity        2
dtype: int64
```

```
df.shape
```

```
(99, 10)
```

```
df.columns
```

```
Index(['Order', 'CompanyId', 'Gender', 'Monthly Income', 'Population',
       'Price', 'Rate', 'Discount', 'Profit', 'Quantity'],
      dtype=object)
```

```
df.dtypes
```

```
Order          int64
CompanyId      int64
Gender         object
Monthly Income int64
Population     int64
Price          int64
Rate           object
Discount       int64
Profit         int64
Quantity       object
dtype: int64
```

```
df.drop(['Order', 'CompanyId'], axis=1)
```

```
df.drop(['Order', 'CompanyId'], axis=1, inplace=True)
```

```
df.head(10)
```

```
df.tail(10)
```

```
df.sample(10)
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df.sample(10).T
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